COMPUTER SYSTEM

INTRODUCTION

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

History

Generation of Computers

The Computer System

- Input
- CPU
- Storage
- Output

HISTORY

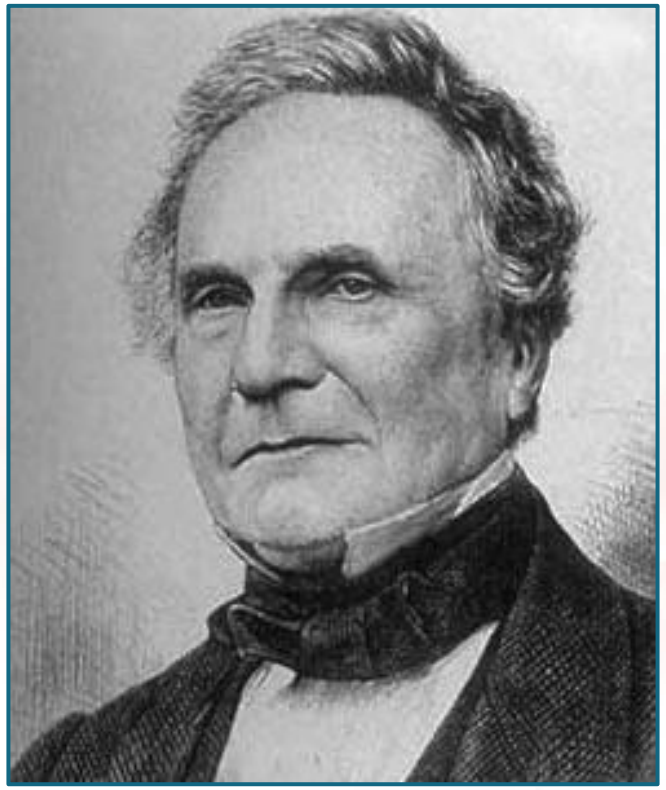
ABACUS

Ultimate **ancestor** of today's computers

The **arithmetic tool** of early merchants

Numerical/logical computations

FATHER OF COMPUTING



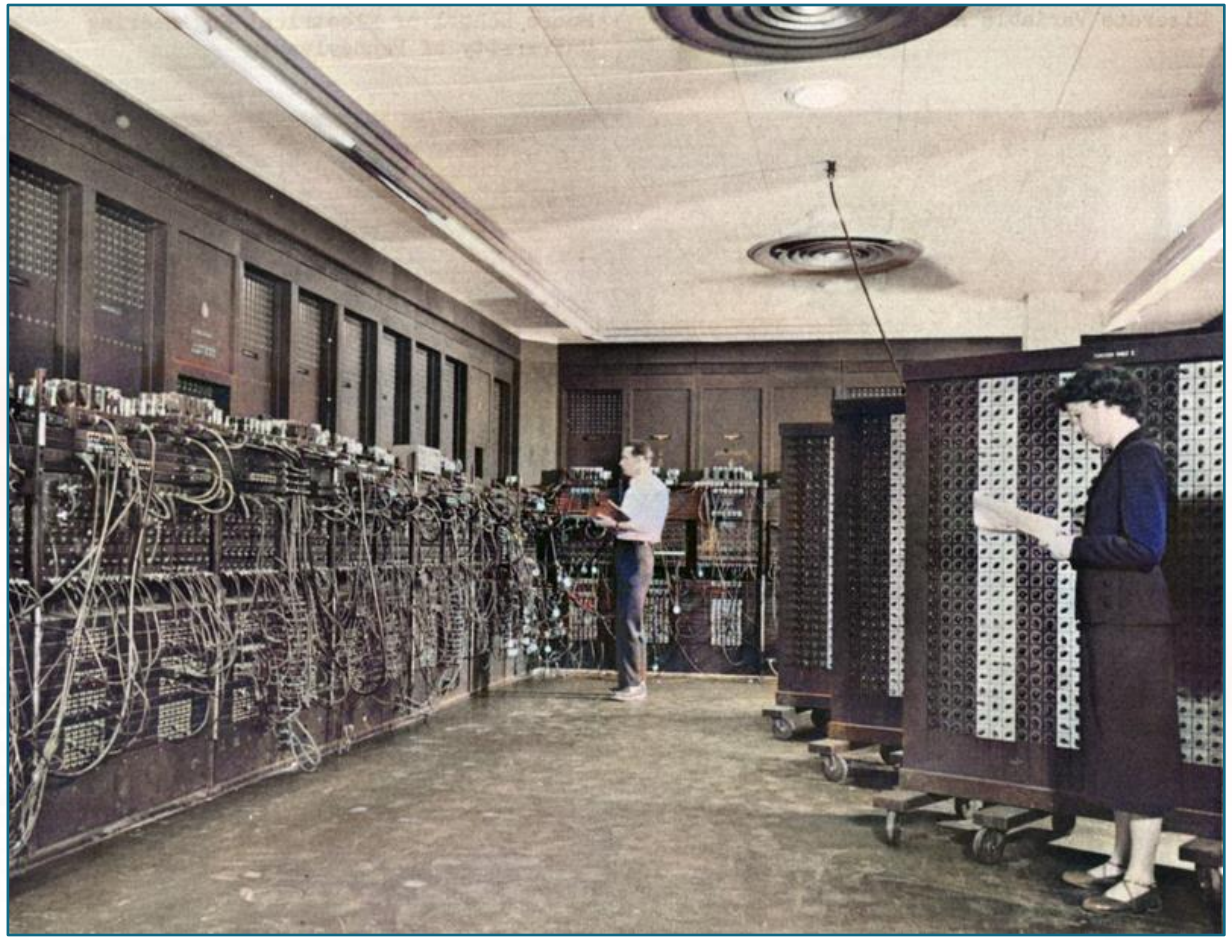
Charles Babbage

Invented the **Analytical** and **Difference Engines**

Embodied the basic **concepts** and **elements** of a modern general-purpose **computer**

GENERATION OF COMPUTERS

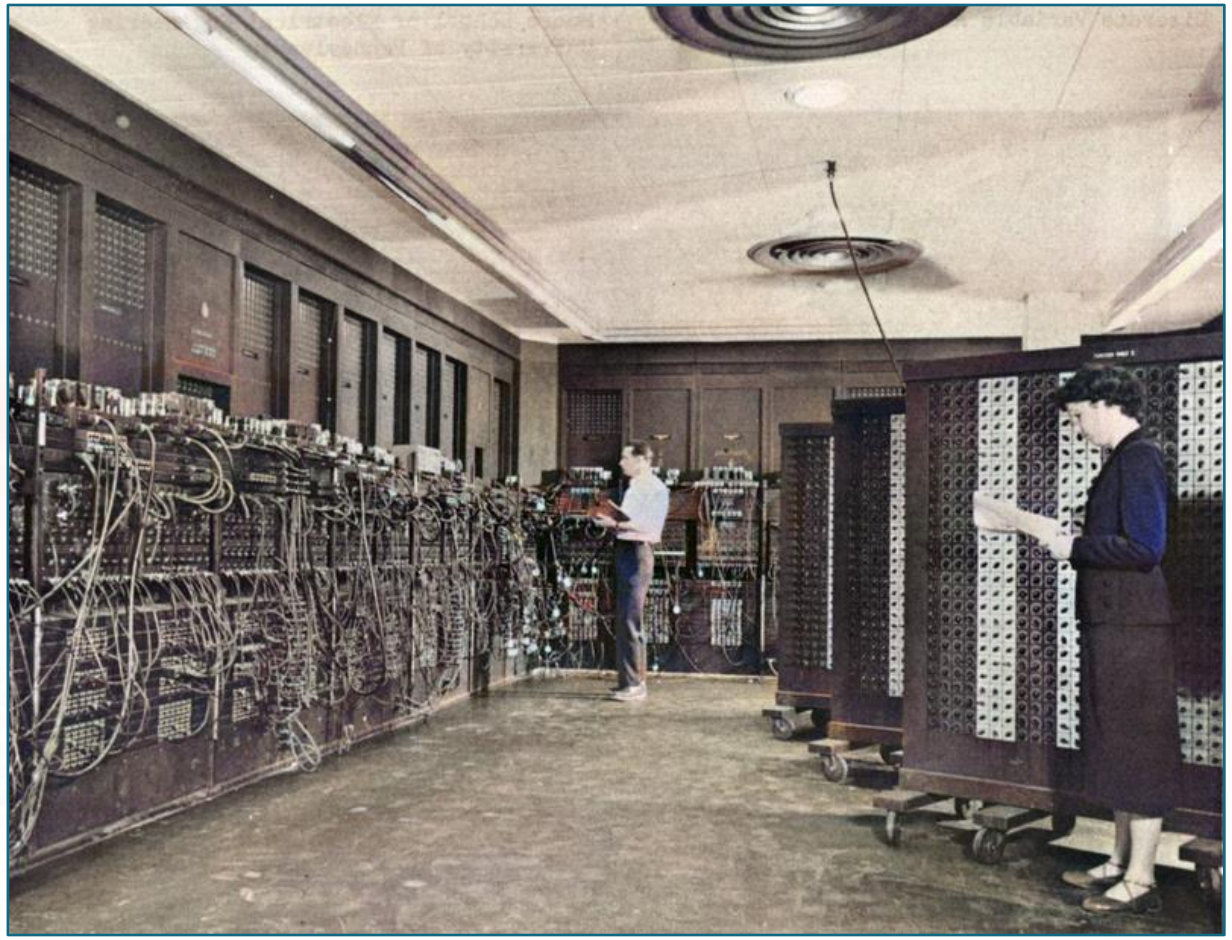
THE FIRST GENERATION



Electronic Numerical Integrator and Computer (ENIAC)

Electronic Discrete Variable Automatic Computer (EDVAC)

THE FIRST GENERATION



Wired circuits containing **vacuum tubes**

Punch cards are the main storage medium

Designed for a **specific task**

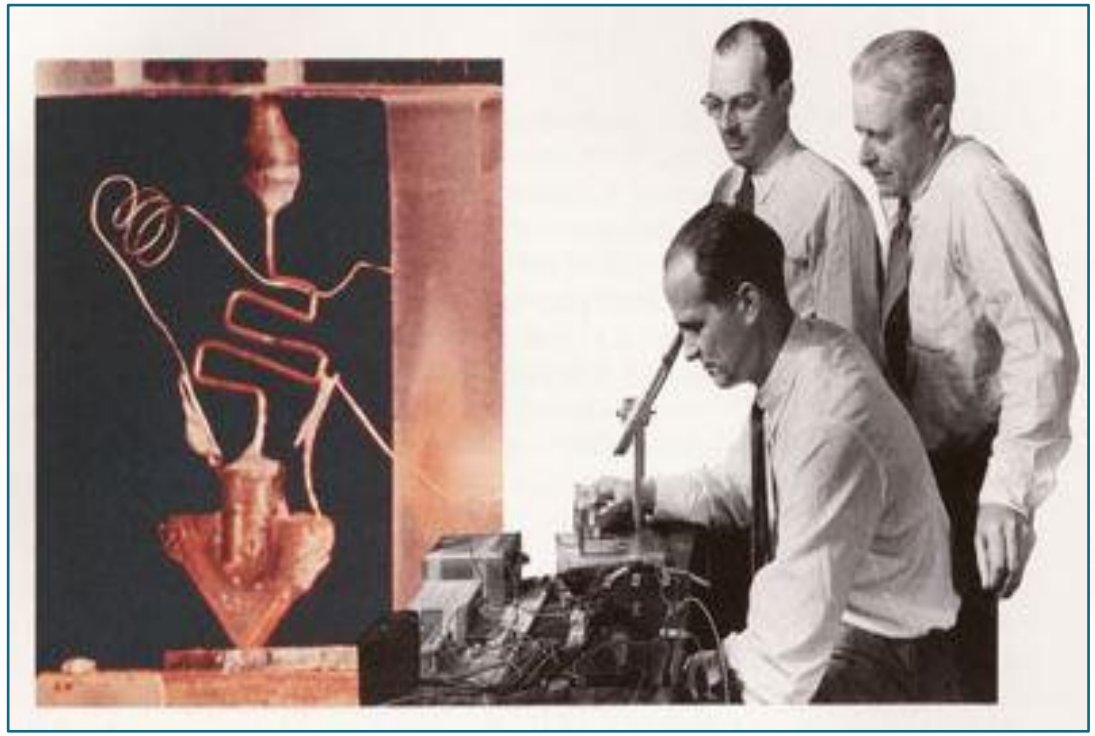
Uses binary codes or "**machine language**"

THE SECOND GENERATION

Transistor-based

Uses **assembly language**

THE SECOND GENERATION

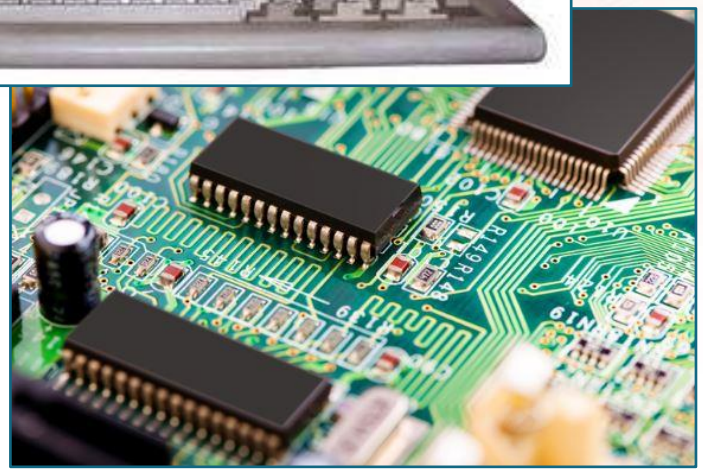


William Shockley

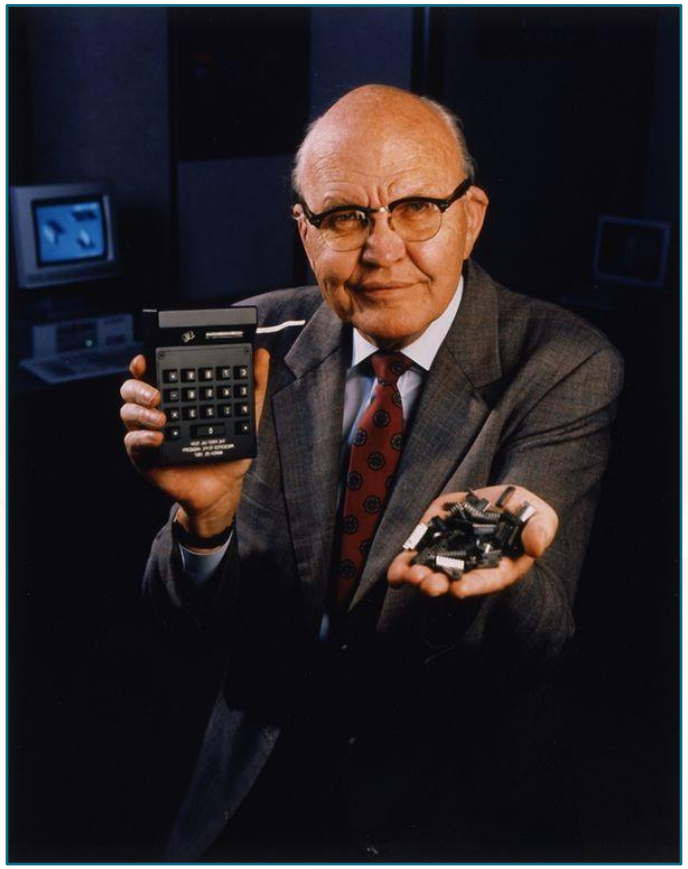
John Bardeen

Walter Houser Brattain

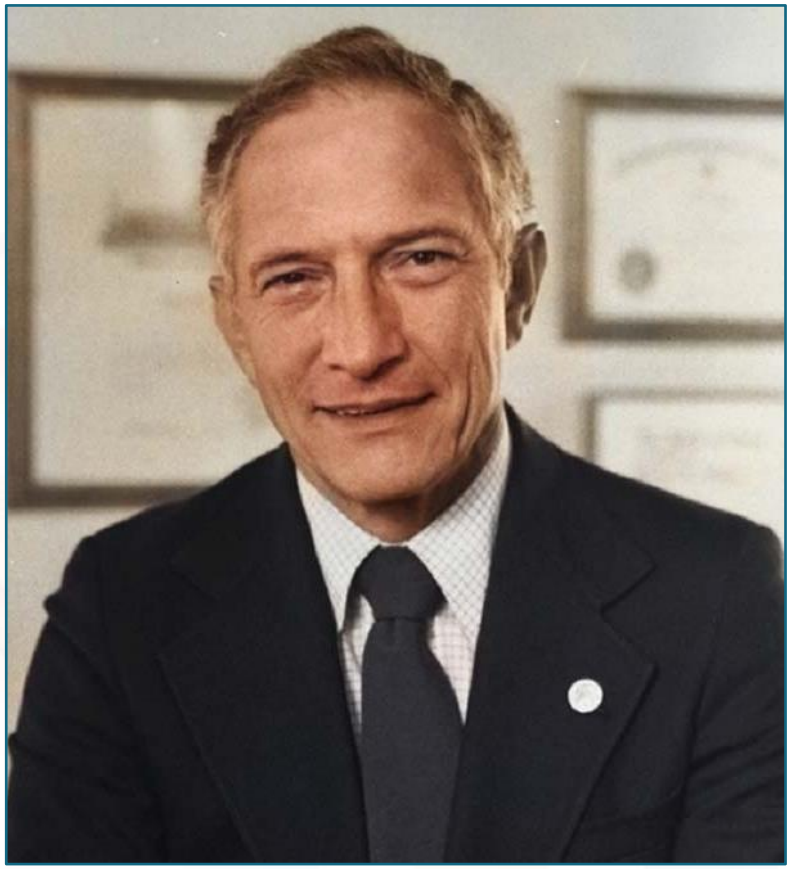
THE THIRD GENERATION



Integrated Circuit (IC) - based



Jack Kilby



Robert Noyce

THE FOURTH GENERATION

Development of user-friendly software packages/**GUI** (Graphical User Interface)

Very Large Scale Integration (**VLSI**)

Ultra Large Scale Integration (**ULSI**)

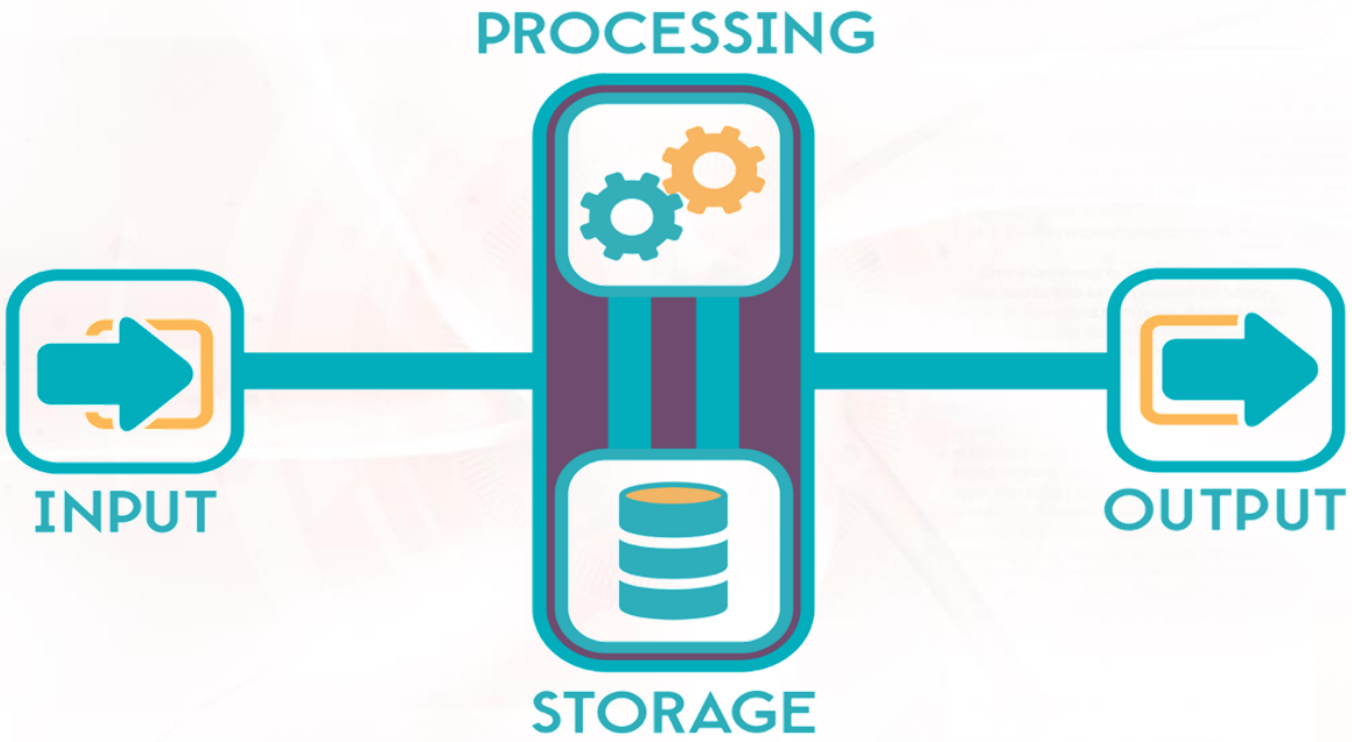
THE FIFTH GENERATION

Artificial Intelligence (**A.I.**)

Natural Language Processing (**NLP**)

THE COMPUTER SYSTEM

THE 4 THINGS COMMON TO ALL COMPUTERS



INPUT

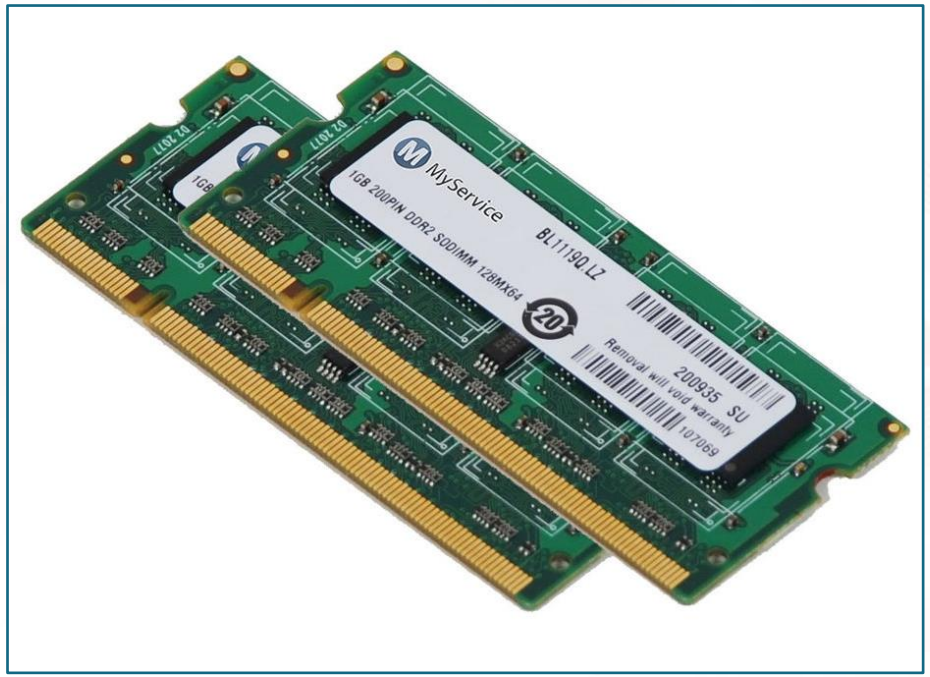
a piece of computer hardware used to **provide data** and control signal to a computer

CENTRAL PROCESSING UNIT (CPU)

Brain of the computer

Controls what the computer does

- calculations
- data processing
- movement of data to and from system memory



Temporarily store information that is currently in use by the computer

Volatile Memory – information stored in RAM is deleted as soon as the computer is turned off

Random Access Memory (RAM)

Read Only Memory (ROM)

Permanently store instructions that tell the computer how to boot (start-up)

Loads the operating system (i.e. Windows)

BIOS (Basic Input/Output System)

Nonvolatile Memory – stored information is not lost when the computer loses power

SECONDARY STORAGE

Permanently store data for long period

High-capacity but **low-speed** data storage

OUTPUT

A peripheral that **receives data** from a computer, usually for display, projection, or physical reproduction

LABORATORY